

MA3236 NONLINEAR PROGRAMMING

Semester 1, AY 2026/2027

Tutorial 2

Review

L03: Convex sets and convex functions

- A set $C \subseteq \mathbf{R}^n$ is **convex** if

$$\mathbf{x}, \mathbf{y} \in C, \lambda \in [0, 1] \implies \lambda \mathbf{x} + (1 - \lambda) \mathbf{y} \in C.$$

Arbitrary intersections preserve convexity; unions generally do not.

- Let C be convex. A function $f : C \rightarrow \mathbf{R}$ is **convex** if

$$f(\lambda \mathbf{x} + (1 - \lambda) \mathbf{y}) \leq \lambda f(\mathbf{x}) + (1 - \lambda) f(\mathbf{y}).$$

It is **strictly convex** when the inequality is strict for $\mathbf{x} \neq \mathbf{y}$ and $\lambda \in (0, 1)$. Concavity and strict concavity are defined by reversing the inequality; equivalently, f is concave iff $-f$ is convex.

- **Jensen's inequality:** if $\lambda_i \geq 0$ and $\sum_{i=1}^m \lambda_i = 1$, then

$$f\left(\sum_{i=1}^m \lambda_i \mathbf{x}_i\right) \leq \sum_{i=1}^m \lambda_i f(\mathbf{x}_i)$$

for a convex f and $\mathbf{x}_i \in C$.

- **Convexity-preserving operations:** nonnegative weighted sums, finite pointwise maxima. A positive weight on a strictly convex summand makes such a weighted sum strictly convex. Pointwise minima generally does not.
- **Composition rule:** if h is convex, then $g \circ h$ is convex when g is convex and nondecreasing, and is concave when g is concave and nonincreasing (on an interval containing $h(C)$). The stated monotonicity is essential.
- The **epigraph** $\text{epi } f = \{(\mathbf{x}, t) : \mathbf{x} \in C, f(\mathbf{x}) \leq t\}$ is convex iff f is convex. If f is convex, every sublevel set $L_\alpha(f) = \{\mathbf{x} \in C : f(\mathbf{x}) \leq \alpha\}$ is convex; the converse need not hold.

L04: First-derivative tools

- In minimizing a convex function over a nonempty convex set, **every local minimizer is global**. Strict convexity makes a minimizer **unique** if it exists; it does not by itself guarantee existence.
- The **gradient** is

$$\nabla f(\mathbf{x}) = \left[\frac{\partial f}{\partial x_1}(\mathbf{x}), \dots, \frac{\partial f}{\partial x_n}(\mathbf{x}) \right]^T \in \mathbf{R}^n.$$

At an interior point where f is differentiable,

$$Df(\mathbf{x}; \mathbf{d}) = \nabla f(\mathbf{x})^T \mathbf{d}, \quad \left. \frac{d}{dt} f(\gamma(t)) \right|_{t=0} = \nabla f(\mathbf{x})^T \gamma'(0)$$

when $\gamma(0) = \mathbf{x}$. For unit Euclidean directions and $\nabla f(\mathbf{x}) \neq \mathbf{0}$, the steepest ascent/descent directions are $\pm \nabla f(\mathbf{x}) / \|\nabla f(\mathbf{x})\|$; if the gradient is zero, all directional derivatives are zero.

- **Gradient normality:** on a differentiable level set $f(\mathbf{x}) = \alpha$, a nonzero gradient is normal to every tangent direction: $\nabla f(\mathbf{x})^T \gamma'(0) = 0$.

- **First-order characterization of convex functions:** if C is open and convex and $f \in C^1(C)$, then

$$f \text{ is convex on } C \iff f(\mathbf{y}) \geq f(\mathbf{x}) + \nabla f(\mathbf{x})^T(\mathbf{y} - \mathbf{x}) \quad \forall \mathbf{x}, \mathbf{y} \in C.$$

Strict convexity is characterized by strict inequality for $\mathbf{x} \neq \mathbf{y}$.

- **Optimality conditions for convex optimization:** if U is open and convex, $C \subseteq U$ is nonempty and convex, and $f \in C^1(U)$ is convex, then

$$\mathbf{x}^* \text{ minimizes } f \text{ over } C \iff \nabla f(\mathbf{x}^*)^T(\mathbf{x} - \mathbf{x}^*) \geq 0 \quad \forall \mathbf{x} \in C.$$

On an open unconstrained domain this reduces to $\nabla f(\mathbf{x}^*) = \mathbf{0}$; at a boundary minimizer the gradient need not vanish.

L05: Second-order tools

- The **Hessian** is

$$H_f(\mathbf{x}) = \left[\frac{\partial^2 f}{\partial x_i \partial x_j}(\mathbf{x}) \right]_{i,j=1}^n \in \mathbf{R}^{n \times n}.$$

It is symmetric when the mixed second partials are continuous, and $\mathbf{d}^T H_f(\mathbf{x}) \mathbf{d}$ gives the second-order directional curvature.

- **Taylor's expansion:** if $f \in C^2(S)$ on an open set containing the segment $[\mathbf{x}, \mathbf{y}]$, then for some $\mathbf{w} \in [\mathbf{x}, \mathbf{y}]$,

$$f(\mathbf{y}) = f(\mathbf{x}) + \nabla f(\mathbf{x})^T(\mathbf{y} - \mathbf{x}) + \frac{1}{2}(\mathbf{y} - \mathbf{x})^T H_f(\mathbf{w})(\mathbf{y} - \mathbf{x}).$$

Equivalently, the local quadratic model is $f(\mathbf{x} + \mathbf{d}) = f(\mathbf{x}) + \nabla f(\mathbf{x})^T \mathbf{d} + \frac{1}{2} \mathbf{d}^T H_f(\mathbf{x}) \mathbf{d} + o(\|\mathbf{d}\|^2)$.

- **P(S)D, N(S)D matrices:** for real symmetric A , PD means $\mathbf{z}^T A \mathbf{z} > 0$ for every $\mathbf{z} \neq \mathbf{0}$, while PSD means $\mathbf{z}^T A \mathbf{z} \geq 0$ for every $\mathbf{z}$. ND and NSD reverse these inequalities; A is indefinite if the quadratic form takes both signs. Equivalently,

$$\begin{aligned} \text{PD} &\iff \lambda_i > 0 \quad \forall i & \text{PSD} &\iff \lambda_i \geq 0 \quad \forall i \\ \text{ND} &\iff \lambda_i < 0 \quad \forall i & \text{NSD} &\iff \lambda_i \leq 0 \quad \forall i \end{aligned}$$

Indefiniteness is equivalent to having both a positive and a negative eigenvalue.

- **Sylvester criterion:** if Δ_k denotes the k th leading principal minor, then

$$A \succ 0 \iff \Delta_k > 0, \quad A \prec 0 \iff (-1)^k \Delta_k > 0 \quad (k = 1, \dots, n).$$

Moreover, $A \succeq 0$ iff *all* principal minors are nonnegative; checking only the leading ones is insufficient for PSD.

- **Second-order characterization of convex functions:** if C is open and convex and $f \in C^2(C)$, then

$$\begin{aligned} f \text{ convex on } C &\iff H_f(\mathbf{x}) \succeq 0 \quad \forall \mathbf{x} \in C, \\ f \text{ concave on } C &\iff H_f(\mathbf{x}) \preceq 0 \quad \forall \mathbf{x} \in C, \\ H_f(\mathbf{x}) \succ 0 \quad \forall \mathbf{x} \in C &\implies f \text{ strictly convex on } C, \\ H_f(\mathbf{x}) \prec 0 \quad \forall \mathbf{x} \in C &\implies f \text{ strictly concave on } C. \end{aligned}$$

An indefinite Hessian at one point implies that f is neither convex nor concave. PD/ND everywhere is sufficient, not necessary, for strictness. The Hessian condition must hold at every point; without the required second derivatives, this test gives no conclusion.

Iff for quadratic functions: for symmetric A and $q(\mathbf{x}) = \frac{1}{2} \mathbf{x}^T A \mathbf{x} + \mathbf{b}^T \mathbf{x} + c$, q is convex iff $A \succeq 0$, strictly convex iff $A \succ 0$, concave iff $A \preceq 0$, and strictly concave iff $A \prec 0$.

1. (a) Prove that the set

$$C = \{\mathbf{x} \in \mathbf{R}^2 : 4x_1^2 + 3x_2 \leq 9, x_2 \geq 0\}$$

is convex.

- (b) Determine whether the set

$$S = \{\mathbf{x} \in \mathbf{R}^3 : x_1 - x_2^2 \leq x_3 \leq x_1 + x_2^2\}$$

is convex. Justify your answer.

Solution. (a) Method 1 (by definition). Let $\mathbf{x}, \mathbf{y} \in C$, let $\lambda \in [0, 1]$, and set

$$\mathbf{w} := \lambda \mathbf{x} + (1 - \lambda) \mathbf{y}.$$

Since $x_2, y_2 \geq 0$, we have $w_2 \geq 0$. Moreover,

$$\begin{aligned} 4w_1^2 + 3w_2 &= 4(\lambda x_1 + (1 - \lambda)y_1)^2 + 3(\lambda x_2 + (1 - \lambda)y_2) \\ &\leq \lambda(4x_1^2 + 3x_2) + (1 - \lambda)(4y_1^2 + 3y_2) \\ &\leq 9, \end{aligned}$$

where we used $2x_1y_1 \leq x_1^2 + y_1^2$. Hence $\mathbf{w} \in C$, so C is convex.

Method 2 (by decomposition). Write

$$C = C_1 \cap C_2,$$

where

$$C_1 = \{\mathbf{x} \in \mathbf{R}^2 : 4x_1^2 + 3x_2 \leq 9\}, \quad C_2 = \{\mathbf{x} \in \mathbf{R}^2 : -x_2 \leq 0\}.$$

The functions $g_1(\mathbf{x}) = 4x_1^2 + 3x_2$ and $g_2(\mathbf{x}) = -x_2$ are convex. Thus C_1 and C_2 are convex sublevel sets by the L03 result, and their intersection C is convex.

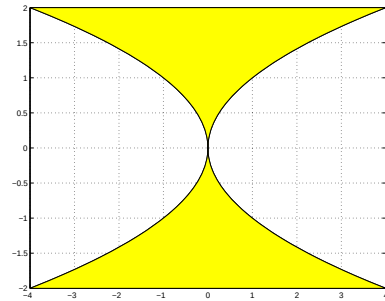
- (b) The set S is not convex. Indeed,

$$\mathbf{u} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} \in S,$$

but

$$\frac{1}{2}\mathbf{u} + \frac{1}{2}\mathbf{v} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \notin S.$$

Thus the midpoint of two points in S can lie outside S .



The slice $x_3 = 0$ in the x_1 - x_2 plane.

2. Assume $A \in \mathbf{R}^{n \times n}$ is symmetric and define

$$f(\mathbf{x}) = \frac{1}{2} \mathbf{x}^T A \mathbf{x}.$$

Determine

(a) $\nabla f(\mathbf{x})$;

(b) $H_f(\mathbf{x})$.

Solution. Write $A = (a_{ij})$. Since

$$f(\mathbf{x}) = \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n a_{ij} x_i x_j$$

and A is symmetric, for each k ,

$$\frac{\partial f(\mathbf{x})}{\partial x_k} = a_{kk} x_k + \sum_{\substack{j=1 \\ j \neq k}}^n a_{kj} x_j.$$

Therefore,

$$\nabla f(\mathbf{x}) = A \mathbf{x}.$$

Differentiating once more gives

$$\frac{\partial^2 f(\mathbf{x})}{\partial x_k \partial x_l} = a_{kl}, \quad k, l = 1, \dots, n,$$

so

$$H_f(\mathbf{x}) = A.$$

3. Let

$$D = \{\mathbf{x} \in \mathbf{R}^2 : x_1 > 0, x_2 > 0\}.$$

Show that the function $f : D \rightarrow \mathbf{R}$ defined by

$$f(\mathbf{x}) = x_1^2 - 4x_1x_2 + 5x_2^2 - \log(x_1x_2)$$

is strictly convex on D .

Solution. The set $D = (0, \infty)^2$ is open and convex, and $f \in C^2(D)$. Its Hessian is

$$H_f(\mathbf{x}) = \begin{bmatrix} 2 + \frac{1}{x_1^2} & -4 \\ -4 & 10 + \frac{1}{x_2^2} \end{bmatrix}.$$

The leading principal minors satisfy

$$\Delta_1 = 2 + \frac{1}{x_1^2} > 0$$

and

$$\Delta_2 = 4 + \frac{2}{x_2^2} + \frac{10}{x_1^2} + \frac{1}{x_1^2 x_2^2} > 0.$$

Thus $H_f(\mathbf{x}) \succ 0$ for every $\mathbf{x} \in D$ by Sylvester's criterion. The second-order characterization then shows that f is strictly convex on D .

4. Classify each matrix as positive/negative definite, positive/negative semidefinite, or indefinite:

$$(a) \begin{bmatrix} 4 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 7 \end{bmatrix}, \quad (b) \begin{bmatrix} 1 & 0 & 3 \\ 0 & 6 & 0 \\ 3 & 0 & -5 \end{bmatrix}, \quad (c) \begin{bmatrix} -3 & 0 & 1 \\ 0 & -4 & 2 \\ 1 & 2 & -5 \end{bmatrix}.$$

Solution. (a) The eigenvalues are the diagonal entries 4, 3, 7, so the matrix is positive definite.

(b) Let A denote the matrix. With

$$P = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad P^T A P = \begin{bmatrix} 6 & 0 & 0 \\ 0 & 1 & 3 \\ 0 & 3 & -5 \end{bmatrix}.$$

Orthogonal similarity preserves eigenvalues. Hence the eigenvalues of A are 6 together with the eigenvalues of $\begin{bmatrix} 1 & 3 \\ 3 & -5 \end{bmatrix}$, namely

$$6, \quad -2 + 3\sqrt{2}, \quad -2 - 3\sqrt{2}.$$

Since there are both positive and negative eigenvalues, the matrix is indefinite.

(c) Its leading principal minors are

$$\Delta_1 = -3, \quad \Delta_2 = 12, \quad \Delta_3 = -44.$$

Thus $(-1)^k \Delta_k > 0$ for $k = 1, 2, 3$. By Sylvester's criterion, the matrix is negative definite.